Multimedia data security

Giulia Boato, Andrea Montibeller, Stefano Dell'Anna

Who am I?

Giulia Boato

Full Professor at MMLab https://mmlab.disi.unitn.it/ leading the Media Forensics Group
Co-funder of Truebees hi-tech start up for deepfake detection

Short bio

MS in Maths,
University of Trento, 2002

PhD in Engineering and Computer Science,
University of Trento, 2005

Visiting researcher at Vigo (Spain), Delft (Nederlands), Tampere (Finland), Innsbruck (Austria)

PI of many EU, national and international projects on multimedia forensics

Schedule

- Monday 8.30-10.30 room A110
- Wednesday 11.30-13.30 room A224

- Esse3 - didattica on line – moodle
- giulia.boato@unitn.it
- andrea.montibeller@unitn.it
- stefano.dellanna@unitn.it

Course outline

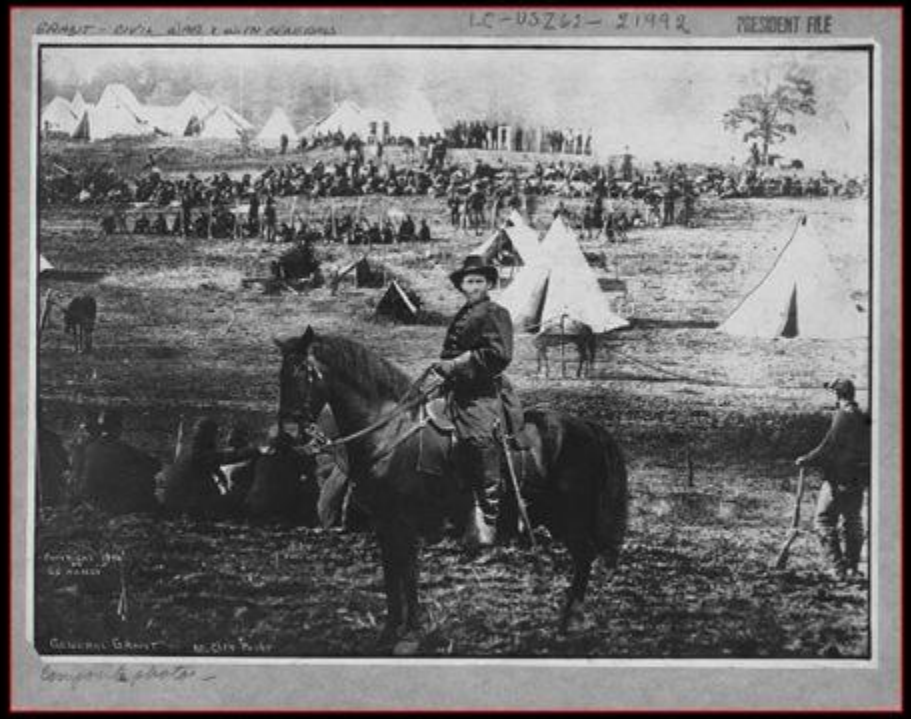
- **Introduction**
  - Motivation & Applications & History
- **Active Forensics**
  - Techniques of digital watermarking
  - Labs and group work
  - Catch the mark 2025! Tbd: my prop is Nov 3
- **Passive forensics**
  - Source identification
  - Tampering detection
  - AI-generated versus real
  - Labs and workshop Euregio Mobility Project 2025

Winner will be invited to present their idea during the workshop of the Euregio Mobility Project 2025 in December

Seeing is not believing

EARLY PHOTOGRAPH

Seeing is not believing



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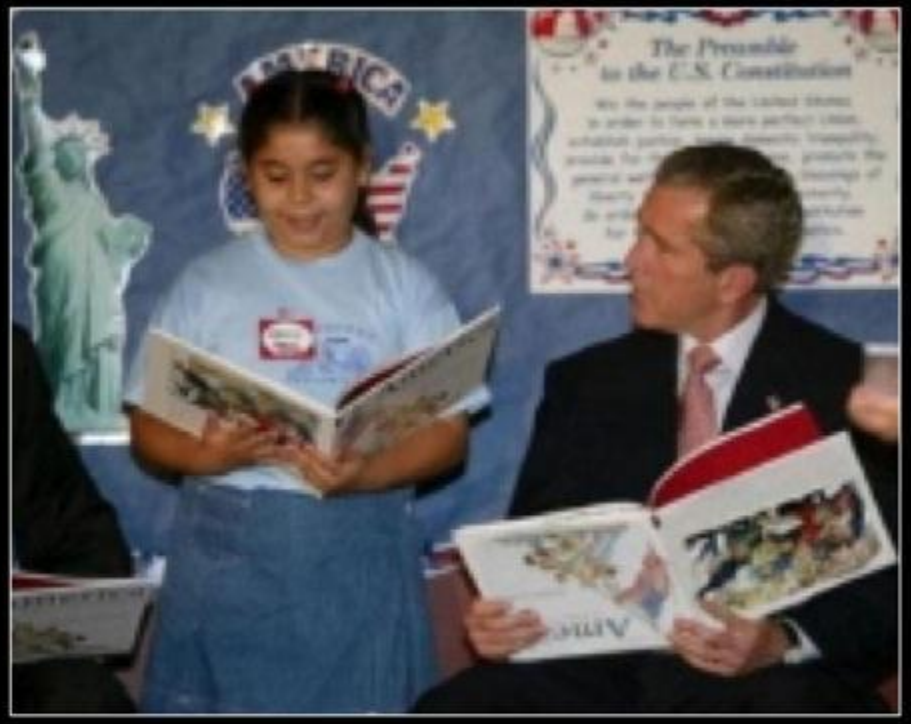

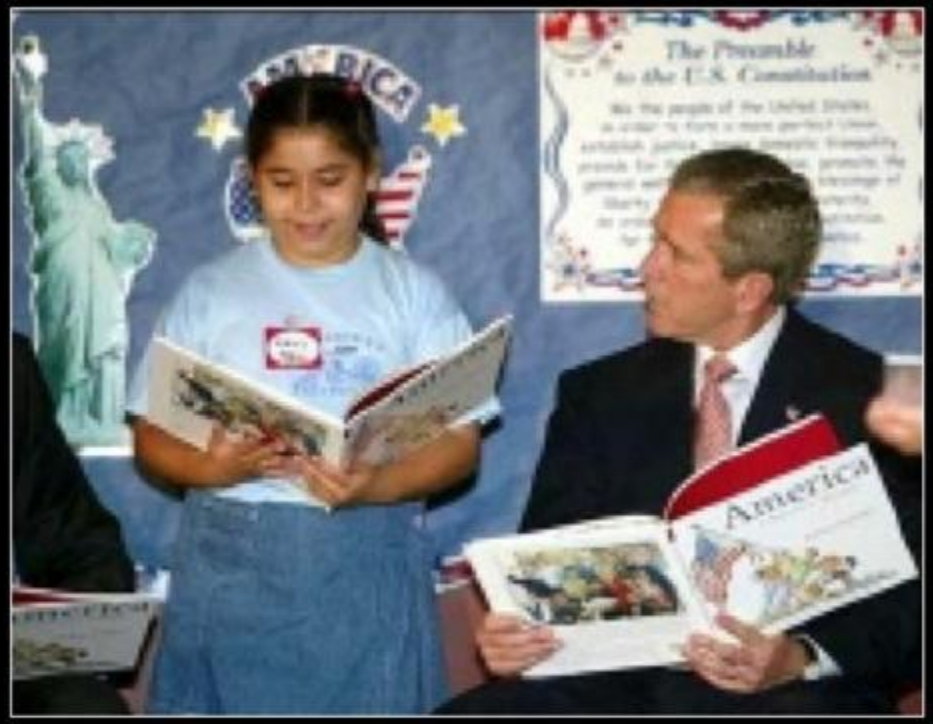

POLITICS

Seeing is not believing

POLITICS

Seeing is not believing

MAGAZINES

Seeing is not believing

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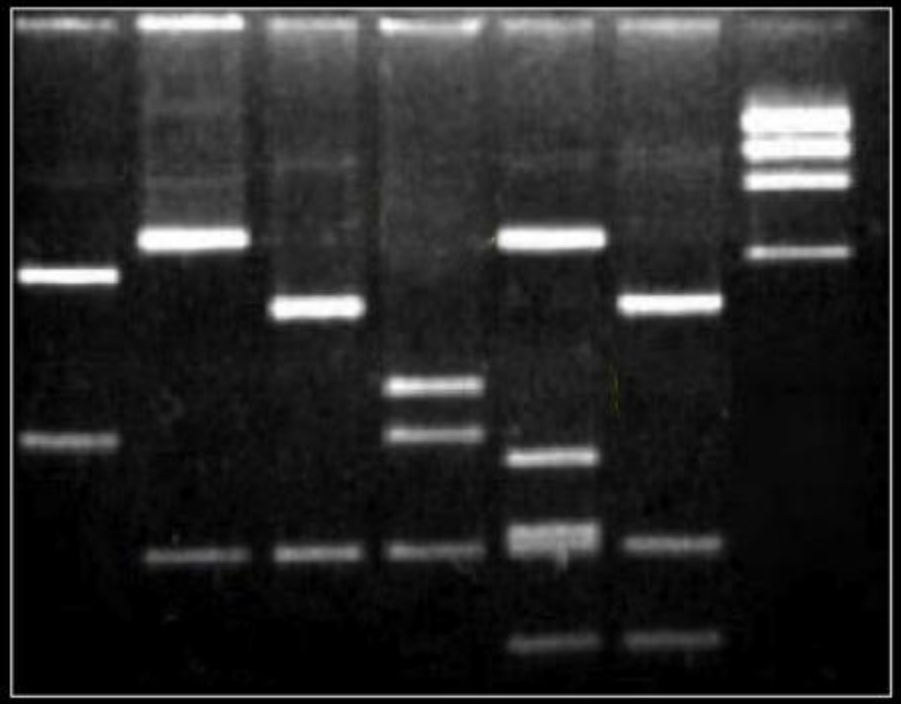
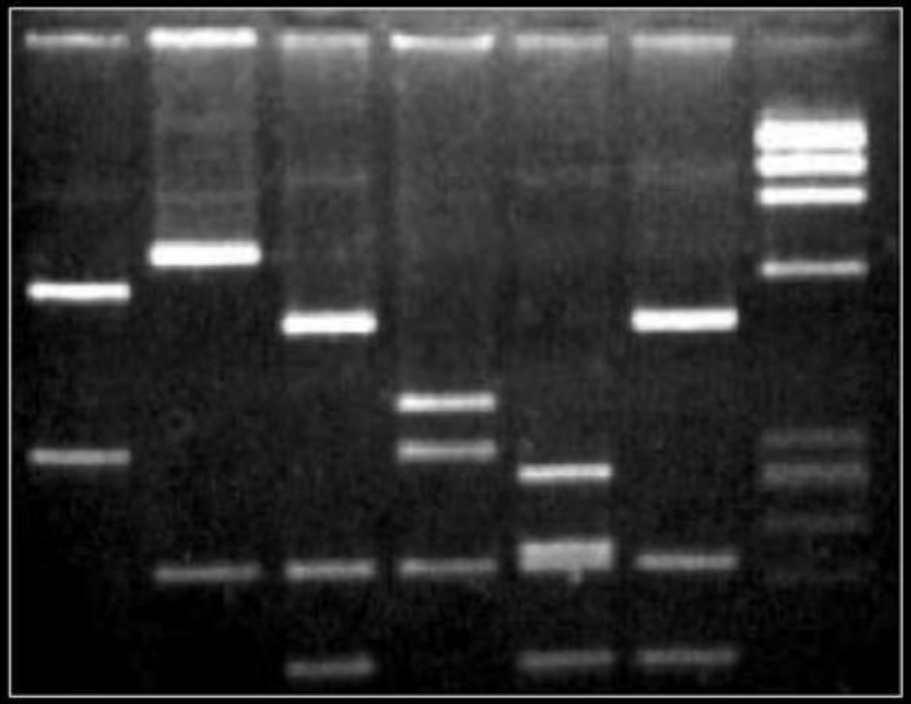
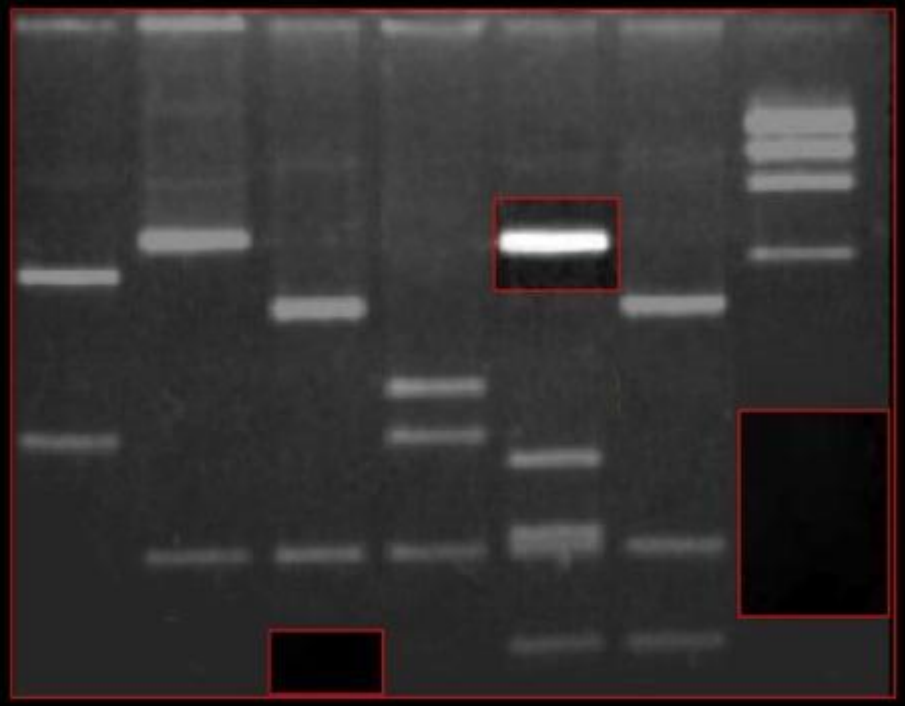




SCIENCE

Seeing is not believing



SCIENCE

Seeing is not believing

NATIONAL SECURITY

Seeing is not believing

NATIONAL SECURITY

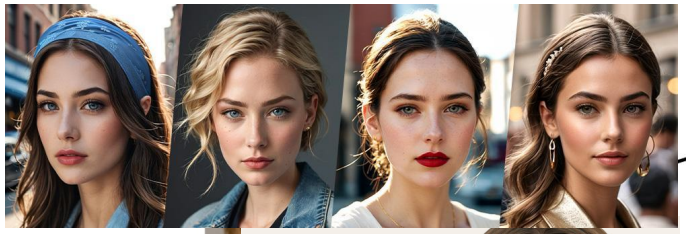
Seeing is not believing

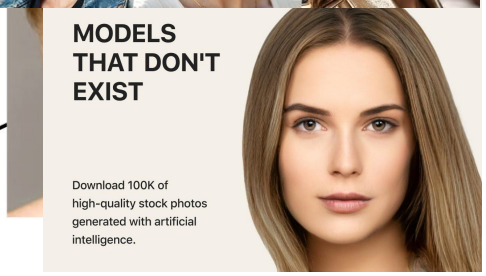
VIDEO

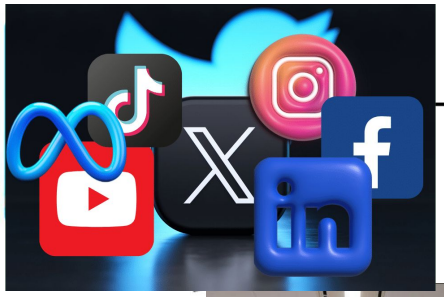
Seeing is not believing

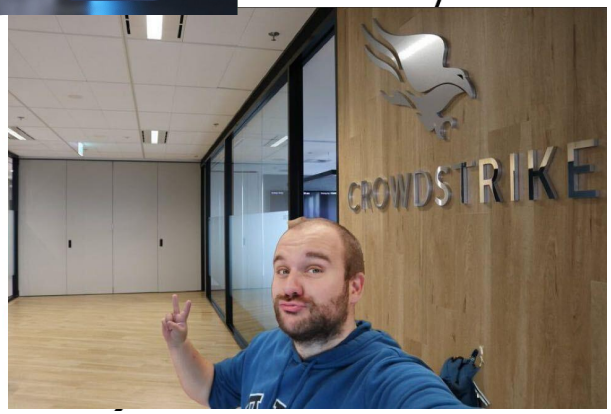
VIDEO

Multimedia forensics - why









Fast and **large-scale** propagation
of modified multimedia content,
potentially carrying a distorted semantic message

Fake News on X (July 19, 2024) about Viral Stuntman, Trainer and Teacher in digital citizenship specialized in Fake News and social networks!

Multimedia forensics - why

- Information trustworthiness
  - **modified or AI-generated data may influence people opinions** and even alter their attitudes in response to the represented event

This is a fake generated by us with FLUX.1 to show how powerfull are new tools!

  - it is important to be able to automatically verify the **authenticity** and the **integrity** of digital images in order to guarantee their truthfulness.

The AI Arms Race to Combat Fake Images Is Even—For Now
Detectors can spot fakes, but generative AI is becoming more subtle
iEEE Spectrum, Jun 2024

Multimedia data security

- What's that?
- Any idea?
- What do you expect?
- What would you like to learn?

Multimedia data security

Fake Obama created using AI video tool - BBC News

https://www.youtube.com/watch?v=AmUC4m6w1wo

It's Getting Harder to Spot a Deep Fake Video

https://www.youtube.com/watch?v=gLoI9hAX9dw

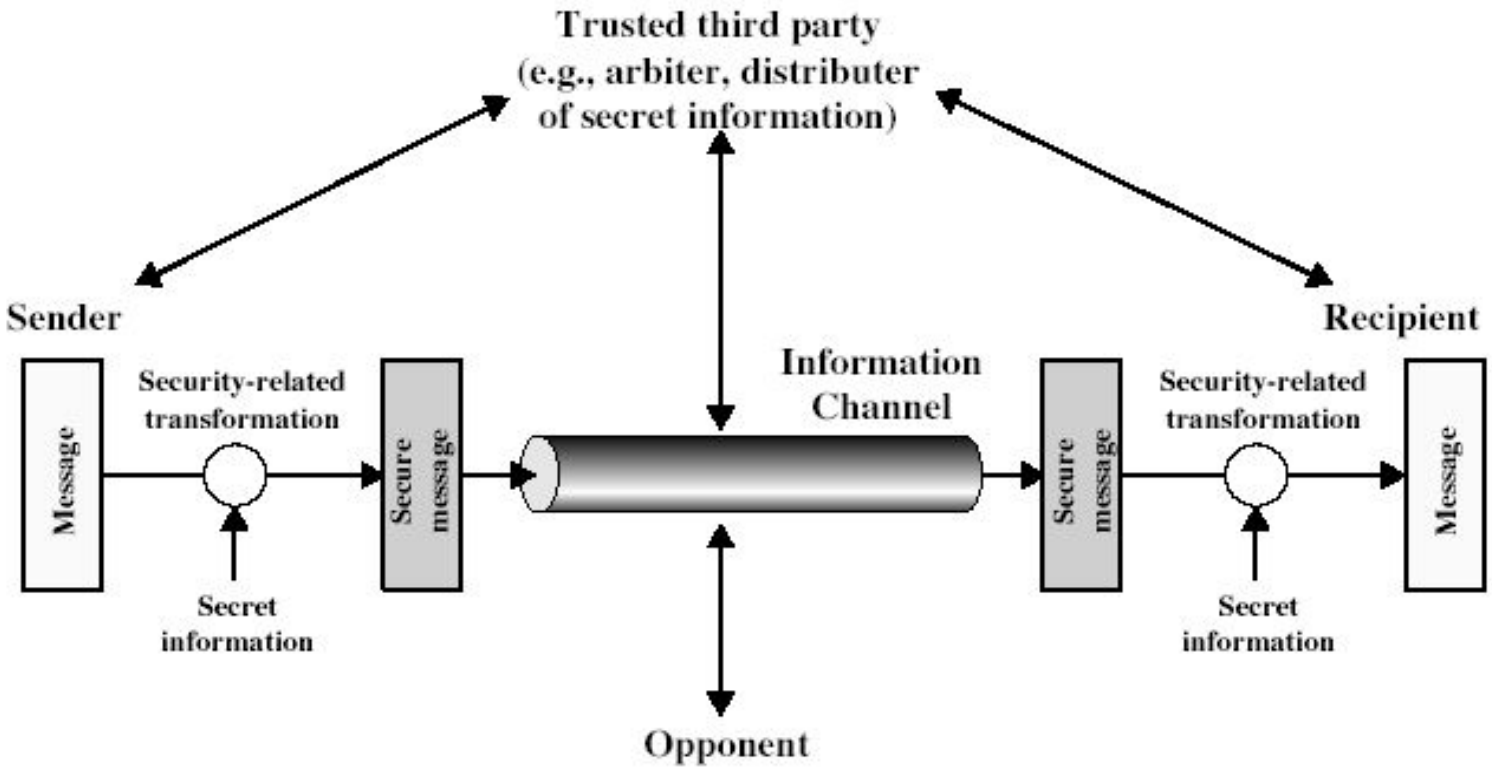
Introduction

- Definition: process of secretly embedding information inside a data source
- Multidisciplinary field
  - multimedia signal processing
  - communication theory
  - information theory and coding theory
  - theory of human perceptual systems
  - law, multimedia industry, consumer electronics

Introduction

- Dissemination of digital documents
- Owner identification
- Forgery detection
- Identification of illegal copies
- Intellectual protection

Introduction



Introduction

- Media items
  - Images
  - Video
  - Audio
  - Graphics
  - Documents
  - Computer programs

Motivation

- Access control
  - control of system access and access limitations to system functionality and data
- Authentication
  - proof of identity of originator/author and of authenticity of data
- Confidentiality
  - avoid that unauthorized third party access data
- Integrity
  - proof that data are not modified by third party
- Evidence
  - check authenticity and integrity of data, also by authorized third party, to guarantee liability of connection

Motivation

- Copyright protection
  - The rapid revolution in digital multimedia and the ease of generating identical and unauthorized digital data. Copy with a few mouse click easy to copy without degradation
  - non-authorized copying must be avoided
  - ownership verification
- Authentication
  - Seeing is believing? easy to modify and now also to generate!!

➡ need to establish reliable methods for copyright protection and authentication

Seeing is believing?

CTV News - 'Bored' horse climbs on top of New Brunswick family's roof

Motivation

- The need to establish secure invisible channels for covert communications
- Adding caption and other additional information without an additional channel
- Learning how to extract information about the data history

Applications

- Watermarking – the practice of imperceptibly altering a work to embed a message about that work
- Fingerprinting – insert different marks into different copies of the same document
- Steganography – (covered writing) the process of secretly embedding information into a data source in such a way its very existence is concealed
- Cryptography – rendering messages unintelligible to any unauthorized persons who might intercept them
- Digital forensics – automatic analysis of data source and possible tampering

Applications

Watermarking

Fingerprinting

Steganography

Cryptography

Active forensics

Digital forensics

Passive forensics

Cryptography

- Cryptography is the art, or science, of keeping messages secret
- It is used to protect sensitive information transmitted over a public channel
- It deals with all aspects of secure messaging, authentication, digital signatures, electronic money…

Cryptography

Once a plaintext has been encrypted, it can not be read without a proper key which is only possessed by authorized listeners. The ciphertext is transmitted on the public, non-secure, channel.

Cryptography

- Encryption
  - Helps to protect confidentiality
  - Protection vanishes after decryption
  - Prefer a way to associate copyright info with MM source even after decryption/compression/transmission/etc.

- Digital cryptographic signature
  - Helps to authenticate sender's identity and data integrity
  - Need to attach a separate signature to the data source
  - Audio/image/video allows imperceptible changes
  - Opportunities for new and seamless ways of authentication

Watermarking

- Encryption does not solve the problem of unauthorized copying
- Multimedia data should be marked to allow distribution to be tracked
- Digital watermarking can provide
  - An additional layer of protection after decryption
  - Data authentication and integrity

Watermarking

- A digital watermark is an identification code bearing information (logo's, signatures) about the copyright owner, authorized consumers and so on.
- It is permanently embedded into digital data for copyright protection, data authentication, integrity checking.
- In most applications the watermark is not visible (perceivable) to a human observer, so that data quality is not degraded.

Watermarking

- Copyright Protection
  - A digital watermark embedded within the host signal can be retreived later to assert the owner's copyright over the marked media
  - Watermark can prove ownership in court when someone has infringed on copyrights
- Copy Protection
  - The information stored in a watermark can directly control digital recording devices for copy protection purposes
  - A watermark can prevent illegal regeneration of copies of the marked data
  - The watermark represents a copy-prohibit bit and watermark detectors in the recorder determine whether the data offered to the recorder may be stored or not

Watermarking

- Broadcasting Monitoring
  - Commercial advertisements and other broadcast signals could carry tracking information in the form of hidden watermarks to monitor and verify the number of times the data has been broadcast. Hence, the customer can be charged accordingly

- Data Authentication
  - Fragile watermarks can be used to check the authenticity of the data and therefore to detect and highlight unauthorized modification to the protected data. These are weak watermarks and are designed to be destroyed in case of alteration of the marked data in an unauthorized manner

Watermarking

- Annotation and indexing
  - Watermarks can also be used to annotate and index digital data. These watermarks can be used by individuals as identifiers leading to the source of the marked data or by search engines to return the marked data in relevant web searches
- Medical applications
  - The date and the patient's details can be embedded as a watermark within medical images, which can be extracted only by authorized parties

Fingerprinting

- Fingerprinting
  - Different watermarks in the copies of the data that are supplied to different customers (serial number)
  - Copy control: to trace the source of illegal copies
  - Traitor tracing: it enables the intellectual property owner to identify customers who have broken their license agreement by supplying the data to third parties

Steganography

- Steganography is the art-science of communicating hiding the existence of the communication.
- In contrast to cryptography, where the enemy is allowed to intercept and modify messages without being able to violate the security ensured by a cryptosystem, the goal of steganography is to hide messages inside other harmless messages in a way that does not allow any enemy to even detect the presence of a second secret message.

Digital forensics

Identify the device that took a particular multimedia data

Determine possible tampering

Discriminate multimedia objects generated by real cameras, scanners, via CG softwares or AIs

Digital forensics

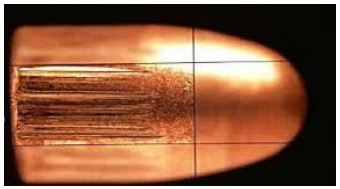
- Was the picture captured using a digital camera? or generated by a CG software? or generated by an AI?
- Which camera brand took this picture? What model?
- What technologies were employed?
- What processing has been done?
- Has it been tampered or manipulated?
- Does it contain hidden data?

Digital forensics

TRUE???

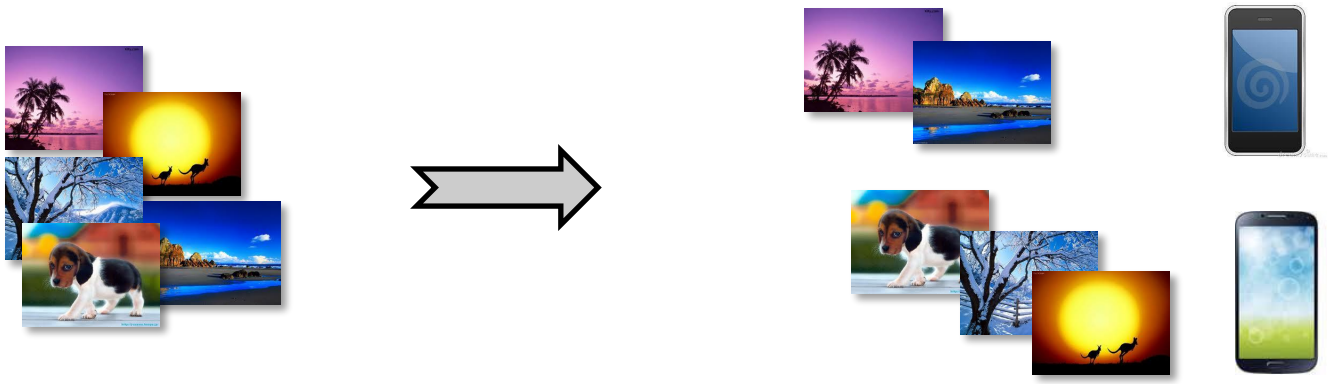
Source identification

Every time a device acquires a multimedia object it leaves imperceptible traces on it



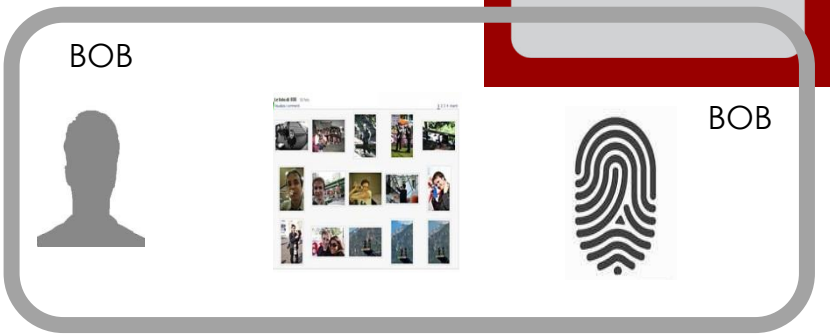
Source identification

In general, **source identification** problem links multimedia content to a particular (class of) acquisition device(s).

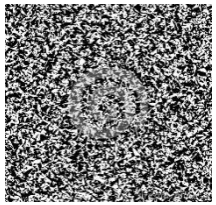


It can be solved at different levels …

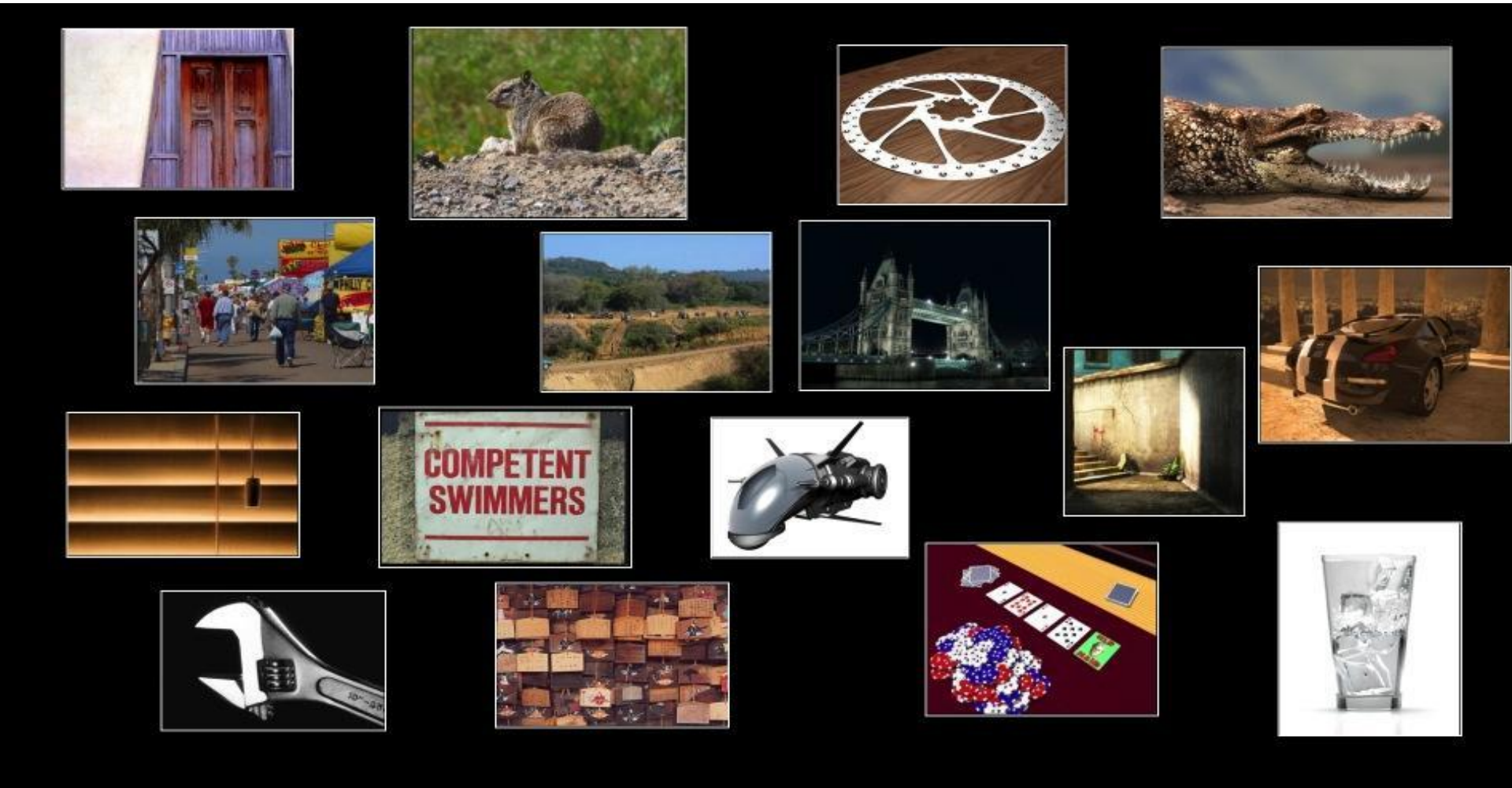
Example of application: photo on Facebook



Who is the author of this image?



Artificial or real?

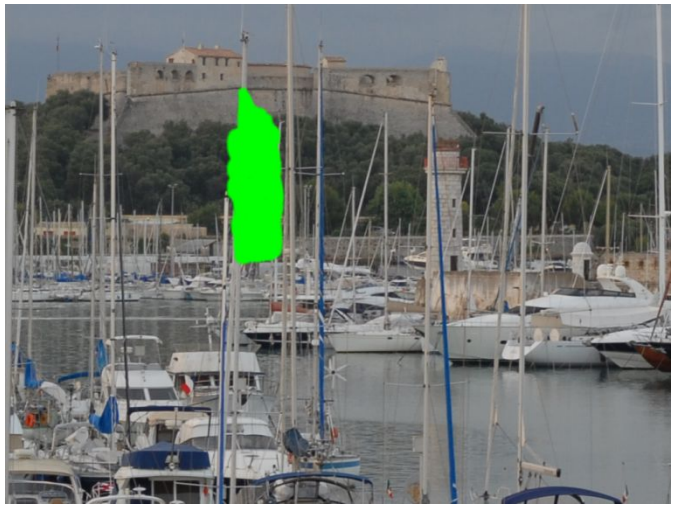


Artificial or real?

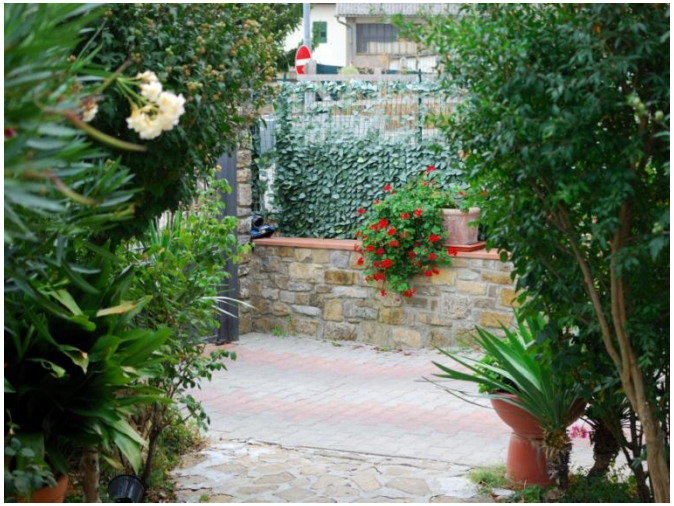
And these data are pretty old…. we will talk about new ones…

Tampering detection

Tampering detection



Tampering detection



QUESTIONS
why
when
what
who
where
how
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